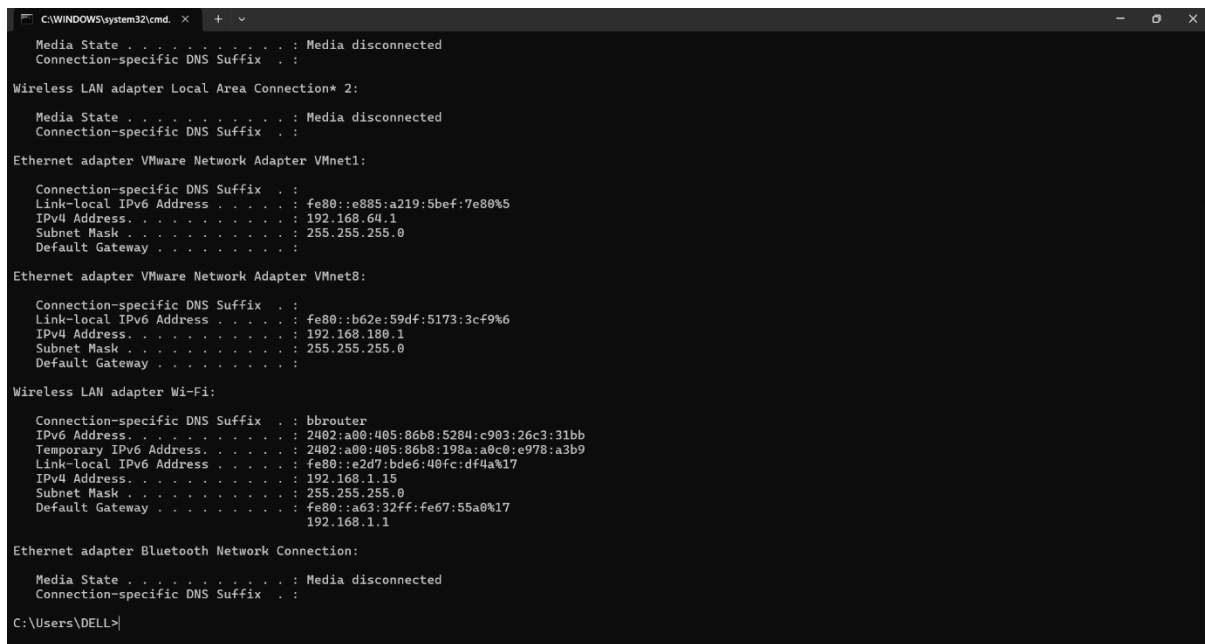


Troubleshooting Tools

Q1. Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.

Ans:

Figure 1: Output of the ipconfig Command

A screenshot of a Windows Command Prompt window titled 'C:\WINDOWS\system32\cmd.' showing the output of the 'ipconfig' command. The output lists network configuration for various adapters: Media State (Media disconnected), Wireless LAN adapter Local Area Connection* 2 (Media disconnected), Ethernet adapter VMware Network Adapter VMnet1 (IPv4: 192.168.64.1, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.1.1), Ethernet adapter VMware Network Adapter VMnet8 (IPv4: 192.168.180.1, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.1.1), Wireless LAN adapter Wi-Fi (IPv4: 192.168.1.15, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.1.1), and Ethernet adapter Bluetooth Network Connection (Media disconnected). The prompt 'C:\Users\DELL>' is visible at the bottom.

```
C:\WINDOWS\system32\cmd.
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Wireless LAN adapter Local Area Connection* 2:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Ethernet adapter VMware Network Adapter VMnet1:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::e895:a219:5bef:7e80%5
IPv4 Address. . . . . : 192.168.64.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
Ethernet adapter VMware Network Adapter VMnet8:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::b62e:59df:5173:3cf9%6
IPv4 Address. . . . . : 192.168.180.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . : bbrouter
IPv6 Address. . . . . : 2402:a00:405:86b8:5284:c903:26c3:31bb
Temporary IPv6 Address. . . . . : 2402:a00:405:86b8:198a:a0c0:e978:a3b9
Link-local IPv6 Address . . . . . : fe80::e2d7:bde6:40fc:df4a%17
IPv4 Address. . . . . : 192.168.1.15
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::a63:32ff:fe67:55a0%17
192.168.1.1
Ethernet adapter Bluetooth Network Connection:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
C:\Users\DELL>
```

Explanation

The **ipconfig** command displays the network configuration of the computer. It helps identify important information such as the **IPv4 address**, **Subnet Mask**, and **Default Gateway**, which are useful for network configuration and troubleshooting.

Observed Values:

IPv4 Address: **192.168.1.15**

Default Gateway: **192.168.1.1**

Q2. Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.

Ans:

Figure 1: Output of the ping Command for www.google.com

```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>ping www.google.com

Pinging www.google.com [2001:4860:482a:7700::] with 32 bytes of data:
Reply from 2001:4860:482a:7700::: time=13ms
Reply from 2001:4860:482a:7700::: time=15ms
Reply from 2001:4860:482a:7700::: time=15ms
Reply from 2001:4860:482a:7700::: time=15ms

Ping statistics for 2001:4860:482a:7700:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 13ms, Maximum = 15ms, Average = 14ms

C:\Users\DELL>
```

Average Response Time: 14ms

Explanation

The **ping** command checks whether the destination server can be reached over the network. It sends small data packets to the server and measures the time taken to receive a reply. A successful response confirms that the network connection is working.

Q3. Run the tracert (or traceroute on Linux/Mac) command to www.flipkart.com and list the number of hops it takes to reach the destination.

Ans: Figure : Output of the tracert Command for www.flipkart.com

```
Command Prompt x + v
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>tracert www.flipkart.com

Tracing route to flipkart.com [103.243.32.90]
over a maximum of 30 hops:

  0  1 ms    1 ms    1 ms  RTK_GW.bbrouter [192.168.1.1]
  1  4 ms    3 ms    2 ms  10.230.192.1
  2  *        *        4 ms  10.111.24.5
  3  12 ms   13 ms   18 ms  136.232.112.109
  4  *        *        *      Request timed out.
  5  *        *        *      Request timed out.
  6  *        *        *      Request timed out.
  7  *        *        *      Request timed out.
  8  *        *        *      Request timed out.
  9  *        *        *      Request timed out.
 10  *        *        *      Request timed out.
 11  37 ms   36 ms   36 ms  103.243.32.90

Trace complete.

C:\Users\DELL>
```

Number of Hops : 11

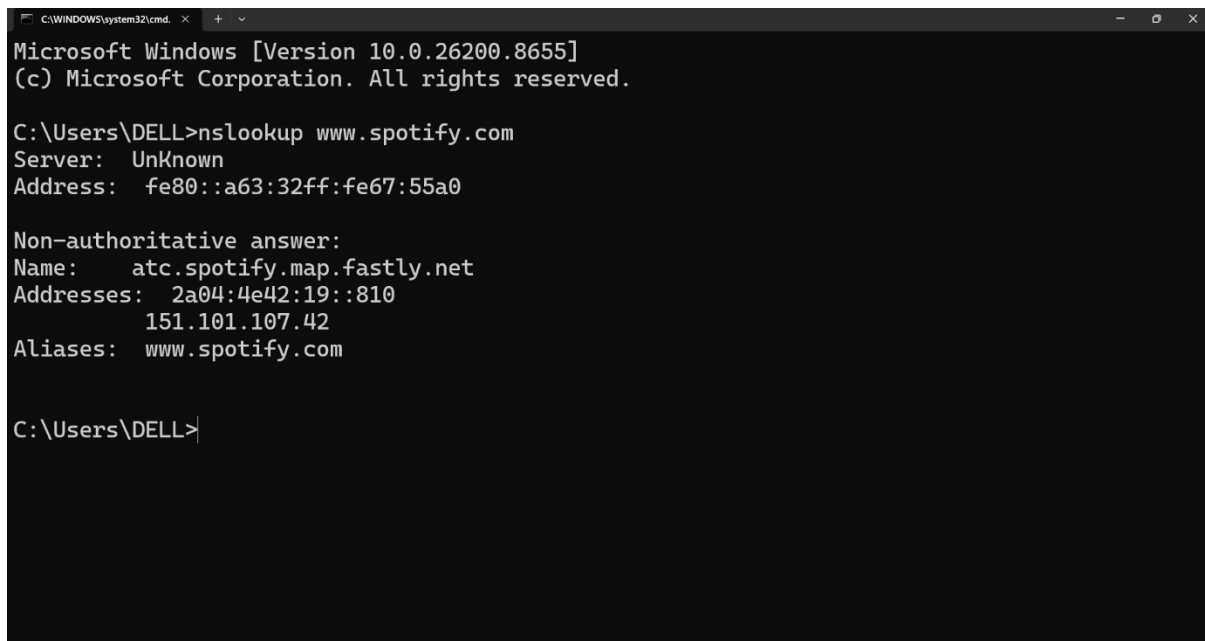
Explanation

The **tracert** command shows the path that data packets travel from the computer to the destination server. Each **hop** represents a router or network device that forwards the packet toward its destination. This command helps identify the route taken and locate possible network delays or connection problems.

Q4. Use the nslookup tool to find the IP address of www.spotify.com and explain in one line what this tool helps you verify.

Ans:

Figure : Output of the nslookup Command for www.spotify.com



```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>nslookup www.spotify.com
Server: Unknown
Address: fe80::a63:32ff:fe67:55a0

Non-authoritative answer:
Name: atc.spotify.map.fastly.net
Addresses: 2a04:4e42:19::810
           151.101.107.42
Aliases: www.spotify.com

C:\Users\DELL>
```

IP Address: 151.101.107.42

One-Line Explanation

The **nslookup** tool verifies whether a domain name is correctly translated into its corresponding **IP address** by the **DNS server**.

Q5. A friend says they cannot access Zomato's website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check. Hint: Think about checking both network connectivity and DNS resolution.

Ans:

Troubleshooting Steps for Accessing Zomato Website:

Step	Command	Purpose
1	ipconfig	Check whether the laptop has a valid IPv4 address and the correct default gateway.
2	ping www.zomato.com	Check whether the laptop can reach the Zomato server over the network.
3	nslookup www.zomato.com	Verify whether the DNS server correctly resolves the Zomato domain name into an IP address.

Explanation

When a user cannot access the Zomato website, the first step is to check the laptop's network configuration using **ipconfig**. Next, use **ping** to test whether the website is reachable. Finally, use **nslookup** to verify that the DNS server is correctly converting the website name into an IP address.

Conclusion

By checking the network configuration, connectivity, and DNS resolution in the correct order, most website access problems can be identified and resolved quickly.